



Basic Principles of Medicine II

Module: Nervous System & Behavioral Sciences

NB IMCQ 10

(Neuroscience and Behavioral Sciences)

1. A 19-year-old man is brought to the emergency department because of decreased responsiveness after falling from a tree 30 minutes ago. Physical examination shows no spontaneous ventilation. He is unresponsive and cannot be aroused by painful or verbal stimuli. His pupils are fixed and dilated, and the vestibular-ocular reflex is absent. Which of the following values best represents the Glasgow coma scale score for this patient?

- A. 1
- B. 3
- C. 6
- D. 8
- E. 12

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- A. 1
- ✓ B. 3
- C. 6
- D. 8
- E. 12

2. A 47-year-old woman is brought to the emergency department because of decreased responsiveness after falling from the roof 30 minutes ago. She is unresponsive to verbal stimuli. Painful vigorous stimuli cause arousal, followed by a return to a sleep-like state. Which of the following terms best describes the level of consciousness in this patient?

- A. Coma
- B. Fully conscious
- C. Lethargic
- D. Obtunded
- E. Stupor

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3. A 19-year-old woman comes to the physician because of a 1-month history of nausea and headaches. Her mother reports that she has become excessively emotional and has displayed personality changes and heightened fearfulness. Neurologic examination shows that fear returns despite repeated safe exposures. MRI of the brain shows a mass. Which of the following lobes is most likely affected in this patient?

- A. Frontal
- B. Insular
- C. Occipital
- D. Parietal
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4. A 42-year-old woman comes to the physician because of a 1-week history of headaches and sudden, irresistible bouts of sleepiness. MRI of the brain shows a lesion. Which of the following regions of the brain is the most likely location of the lesion?

- A. Cerebellum
- B. Lateral hypothalamus
- C. Prefrontal cortex
- D. Thalamus
- E. Ventrolateral preoptic region

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(5) A 25-year-old woman comes to the physician because of a 6-month history of recurrent, unexpected episodes of panic attacks. In the past few months, she has also become increasingly fearful of being in situations where escape might be difficult or help unavailable if she has another episode. As a result, she avoids using public transportation, such as buses and trains, and going to crowded places, including supermarkets. She no longer leaves home alone and asks her friends to run errands for her. Laboratory investigations show thyroid-stimulating hormone (TSH)=3.0 (normal range=0.4–4.0 mIU/L), free thyroxine (T4)=0.85 (normal range=0.7–1.9 ng/dL), and random, non-fasting glucose 110 mg/dL (normal range≤140 mg/dL). A 24-hour urine shows normal metanephrine levels. An electrocardiogram shows a normal sinus rhythm. Vital signs are within the normal range. Physical examination shows no tremor, lid lag, exophthalmos, or thyroid enlargement. Which of the following is the most likely diagnosis?

- A. Hypoglycemia
- B. Generalized anxiety disorder
- C. Pheochromocytoma
- D. Panic disorder
- E. Agoraphobia
- F. Hyperthyroidism
- G. Panic disorder and agoraphobia
- H. Social anxiety disorder

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(6) A 30-year-old woman comes to the physician because of a 3-month history of intense anxiety that affects her job performance. She is a sales representative required to deliver weekly client presentations. However, these presentations cause her to experience palpitations, chest pains, excessive sweating, dizziness, and shortness of breath because she fears being judged for any embarrassing mistakes. Her anxiety is worsening, and she is considering leaving her job. Which of the following is the most appropriate short-term treatment?

- A. Stimulant medication
- B. Psychodynamic therapy
- C. Aversion therapy
- D. Covert desensitization
- E. Benzodiazepine
- F. Beta blockers

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(7) An 18-year-old woman is brought to the physician by her parents because of a 6-month history of depressed mood. Since the sudden death of her boyfriend, she says she no longer experiences any moments of happiness. She has lost interest in hobbies and spends most of the day in bed. She has experienced significant weight loss, early morning awakenings, and low energy. She feels guilty and worthless and is hopeless that her life will improve. She has thoughts of suicide, wanting to end her unrelenting feelings of misery. Which of the following is the most likely diagnosis?

- A. Major depressive disorder
- B. Persistent depressive disorder
- C. Adjustment disorder
- D. Normal grief
- E. Prolonged grief disorder

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(8) A 55-year-old woman comes to the physician for a follow-up examination. Over the past 6 months, she has increasingly withdrawn from church and family gatherings that she previously enjoyed. She has a depressed mood, but whenever her children visit, she feels more positive. She sleeps for more than 10 hours every night, has poor concentration at work, and excessive guilt about past mistakes. Her appetite has increased, leading to recent weight gain, and she describes her legs as feeling heavy. She is increasingly sensitive to rejection and criticism. Her symptoms have been treatment-resistant, and a different class of medication was recently started. Since beginning this treatment, she has had 2 episodes of severe headache and markedly elevated blood pressure shortly after eating pepperoni and cheese pizza. Which of the following is most likely the mechanism of action of the medication started?

- A. Inhibition of serotonin and norepinephrine reuptake with muscarinic, histamine, and α_1 -adrenergic receptor blockade
- B. Inhibition of monoamine oxidase
- C. Selective inhibition of serotonin reuptake
- D. Inhibition of serotonin and norepinephrine reuptake
- E. Blockade of β -adrenergic receptors

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(9) A 23-year-old woman is brought to the physician by her partner because of a 2-week history of unusual behavior. During this time, she has slept 3-4 hours per night yet says she feels energetic. She describes her mood as uncharacteristically elevated and says she feels “better than ever.” She has racing thoughts and speaks rapidly without pausing, making it difficult to interject. Her coworkers say she has become more arrogant and distractible, although her supervisor has praised her for efficiently completing multiple projects. Medical records show she had a 3-month episode of depressed mood, loss of interest, low energy, poor concentration, and feelings of worthlessness during the previous year. Current toxicology screening shows no evidence of substance use. Which of the following is the most likely diagnosis?

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(10) A 25-year-old woman is brought to the physician by her husband because of a 6-month history of behavioral changes. After touching commonly used surfaces such as doorknobs, countertops, or chairs, she becomes overwhelmingly distressed by the belief that she has been dangerously contaminated with germs. She is certain that the contamination cannot be removed unless she washes her hands at least 20 times. She spends several hours every day washing her hands and has developed severe dermatitis. Her constant handwashing has led to her losing her job and difficulty caring for her children. Which of the following is the most likely diagnosis?

- A. Delusional disorder, somatic type
- B. Obsessive-compulsive disorder, absent insight/delusional belief
- C. Unspecified obsessive-compulsive and related disorder
- D. Generalized anxiety disorder
- E. Specific phobia, situational

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(11) A 19-year-old man comes to the physician because of a 6-month history of psychological problems. These problems began after he witnessed a coworker become trapped in industrial machinery and die at work. He has recurrent nightmares about the event and avoids returning to his workplace. He is hypervigilant and easily startled by sudden movements and sounds. The physician is considering a diagnosis of posttraumatic stress disorder. Which of the following additional symptoms is most likely required to fulfill the criteria for this diagnosis?

- A. Refusal to talk about the accident
- B. Dissociative amnesia
- C. Intrusive recollections of his coworker's death
- D. Reckless, self-destructive behavior
- E. Distress when encountering reminders of the workplace accident

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(12) A 45-year-old man comes to the physician for a follow-up examination. He was diagnosed with peptic ulcer disease 7 months ago. Repeat endoscopy shows healing of the ulcer with no evidence of malignancy. He has also undergone additional evaluations, all of which have been normal. However, he reports persistent epigastric pain that he describes as severe and continues to request additional testing. He is unable to concentrate at work because he is preoccupied with the possibility that his symptoms indicate a serious illness. Which of the following is the most likely diagnosis?

- A. Functional neurological symptom disorder (conversion disorder)
- B. Factitious disorder
- C. Illness anxiety disorder
- D. Somatic symptom disorder
- E. Delusional disorder, somatic type
- F. Malingering

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(13) A 22-year-old man comes to the physician because of a 2-year history of excessive daytime sleepiness. He sleeps at least 8 hours per night but still falls asleep irresistibly at work. He periodically falls to the ground when experiencing intense emotion and is unable to move for several seconds despite retaining consciousness. Multiple sleep latency testing shows decreased sleep latency. Which of the following additional findings best confirms the patient's diagnosis?

- A. Increased REM latency
- B. Sleep-onset REM periods
- C. Increased total REM sleep time
- D. Increased total sleep time
- E. Decreased N2 sleep
- F. Increased N3 sleep

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(14) A 20-year-old man is brought to the physician by his parents because of a 9-month history of progressive weight loss. He says he has been restricting his caloric intake to “get leaner and improve muscle definition,” but still thinks he looks “too soft.” He expresses intense fear of gaining weight and becomes anxious when encouraged to eat more. He reports recurrent episodes of eating large amounts of food with loss of control, followed by excessive exercise and fasting. His BMI is 17.5 kg/m². Which of the following is the most likely diagnosis?

- A. Unspecified eating disorder
- B. Anorexia nervosa, restricting type
- C. Anorexia nervosa, binge-eating/purging type
- D. Bulimia nervosa
- E. Binge-eating disorder

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